online vendor. In this case, trust is quantified based on information like your credit rating and other credentials. Thus, building trust between the vendor and customer involves at least one intermediary and maybe more.

The cases I've described are just two of the many possibilities in the trust continuum, from simple peer-to-peer interactions between neighbors to complex financial systems. But how do you resolve the trust in a decentralized system in which no organizations or individuals act as intermediaries? Who or what fills this role in such systems? Blockchain can. It is ideally positioned to provide automated trust intermediation through its innovative infrastructure, unique protocol, and distributed ledger technology. It addresses trust and integrity by verification, validation, and protocol-level consensus, and through its distributed immutable recording.

In this chapter, you'll learn about trust and integrity in the context of blockchain-based decentralized systems. You'll learn to design smart contracts with additional techniques for improving trust in the system you developed in chapter 2. Often, the techniques for enabling trust, such as access control, encryption, and digital signatures, also address the integrity requirements of a system. We'll focus on the access control aspect in this chapter, and explore cryptography and hashing techniques in chapter 5.

The chapter introduces a new decentralized application for balloting in a digital democracy (Ballot), as well as a new design diagram in finite state machines (FSM). This chapter also illustrates the use of Solidity features including modifiers and `require()` and `assert()` declarations to implement verification and validation.